Vector spaces

1. Definition and basic properties:

Def. A vector space V over a field F (an F-vector space) is a set V with an addition $+$ s.t. $(V,+)$ is an abelian group, and a scalar multiplication $\cdot : F \times V \to V$, satisfying:

1. $\alpha(v+w) = \alpha v + \alpha w$
2. $(\alpha+\beta)v = \alpha v + \beta v$
3. $\alpha(\beta v) = (\alpha\beta)v$
4. $1 \cdot v = v$

, $\forall \alpha, \beta \in F$, $v, w \in V$

Obs. Elements of V are called vectors and elements of F are called scalars. The scalar multiplication depends on the choice of F.

Obs. For a vector space V over F:

1. $\alpha \cdot 0_V = 0_V$
2. $0_F \cdot v = 0_V$
3. $\alpha \cdot v = 0_V \Rightarrow \alpha = 0_F$ or $v = 0_V$

ex. 1) $V = \mathbb{C}^n$ has both an $\mathbb{R}$-vector space and a $\mathbb{C}$-v.s.

2) $V = F[x]$, all polynomials with coefficients in F, is an F vector space

3) $M_{m,n}(F)$, with the usual addition and scalar multiplication, is an F vector space.

4) $C_{[a,b]}$, the continuous real functions on $[a,b]$, is an $\mathbb{R}$ vector space

2. Subspaces and direct sums:

Def. Let V be a vector space over F. A subset $U \subset V$ is a

subspace of V if it is stable under the operations of V:

1) $v+u \in U, \forall v,u \in U$
2) $\alpha v \in U, \forall \alpha \in F, v \in U$

and the induced operations satisfy the vector spaces axiom.

Th. Let V be a vector space over F and $U \subset V$ nonempty. U is a subspace of $V \iff$:

$$\left.\begin{array}{l} 1)\ v-u \in U, \forall v,u \in U \\ 2)\ \alpha v \in U, \forall \alpha \in F, v \in U \end{array}\right\} \iff \alpha v + \beta u \in U, \forall \alpha,\beta \in U, \forall v,u \in$$

Def. Let V be a vector space and $U, W \subset V$ two subspaces. The sets: $U \cap W$ and $U+W=\{u+w \mid u \in U, w \in W\}$ are subspaces of V.

Obs. $U \cap W$ is called the intersection subspace and $U+W$ the sum subspace. These definitions extend naturally to any finite family of subspaces.

Def. Let V be a vector space over F and $S \subset F$ nonempty. Then te set $\langle S \rangle$ is a subspace of V:

$$\langle S \rangle = \left\{ \sum_{i=1}^{n} \alpha_i v_i \mid \alpha_i \in F, v_i \in S, n \in \mathbb{N} \right\}$$

Obs. $\langle S \rangle$ is the span (linear hull) of S (the smallest subspace of V containing S):

$$\text{For every subspace } U \supset S \Rightarrow \langle S \rangle \subset U$$

Def. Let $U_1, \dots, U_m \subset V$ be subspaces. The sum $U_1 + \dots + U_m$ is a direct sum, written $U_1 \oplus \dots \oplus U_m$, if every $u \in U_1 + \dots + U_m$ has a unique decomposition $u = u_1 + \dots + u_m$, with $u_i \in U_i$.

Obs. (Case of two subspaces) The sum $U+W$ is direct $\Leftrightarrow U \cap W = \{0_V\}$.

3. Basis and dimension:

Def. A nonempty $S \subset V$ is a system of generators for V if every $v \in V$ is a linear combination of finitely many elements of S. V is finitely generated if some finite S generates it.

Obs. A nonempty $L \subset V$ is linearly independent if, for every finite $\{v_1, \dots, v_n\} \subset L$:

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0 \Rightarrow \alpha_1 = \dots = \alpha_n = 0$$

Otherwise L is linearly dependent.

Def. 1) A subset $B \subset V$ is a basis if it is both a system of generators and linearly independent.

2) Every $v \in V$ has a unique representation as a linear combination of vectors from B.

Th. 1) Every vector space $V \neq \{0\}$ has a basis.

2) Let $V \neq \{0\}$ be finitely generated over F. From every finite system of generators one can extract a basis.

Obs. Let S be a system of generators for V. Every linearly independent $L \subset S$ can be completed to a basis of V.

Th. Let V be finitely generated over F. Every linearly independent system of vectors in V can be completed to a basis of V.

Th. Let V be a finitely generated F-vector space. Every basis of V is finite and all bases have the same number of elements.

Def. Let $V \neq \{0\}$ be finitely generated over F. The number of elements of a basis of V is the **dimension** of V, denoted $\dim_F V$. ($V = \{0\} \Rightarrow \dim_F V = 0$).

Obs. If $\dim_F V = n$, then any set of $m > n$ vectors of V is linearly dependent.

Obs. Let $\dim_F V = n$:

1. Any linearly independent system of n vectors is a basis; any system of $m > n$ vectors is linearly dependent.
2. Any system of n generators is a basis; any system of $m < n$ vectors fails to generate V.

Obs. The dimension of a finite dimensional vector space equals:

1. the number of vectors in a basis.
2. the minimal number of vectors in a system of generators.
3. the maximal number of vectors in a linearly dependent system.

Th. 1) Every linearly independent list of vectors in a finite dimensional vector space can be extended to a basis of the vector space.

2) Let V be a finite dimensional and U a subspace of V. There exists a subspace W of V s.t. $V = U \oplus W$

Th. If U and W are subspaces of a finite dimensional vector space V, then:

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$$

Obs. If $U \cap W = \{0\}$ the sum is direct and the dimension is additive.

Th. Let V be a finite dimensional, $U_1, \dots, U_m$ subspaces of V with:

$$V = U_1 + \dots + U_m \text{ and } \dim V = \dim U_1 + \dots + \dim U_m$$

Then $V = U_1 \oplus \dots \oplus U_m$

Th. Let V be a vector space over F with basis $B = (e_i)$, $i \in I$. For every $v \in V$, $v \neq 0$ there is a unique finite subset $\{e_{i_1}, \dots, e_{i_k}\} \subset B$ and nonzero scalars $\alpha_{i_1}, \dots, \alpha_{i_k}$ s.t.:

$$v = \alpha_{i_1} e_{i_1} + \dots + \alpha_{i_k} e_{i_k}$$

4. Local computations and change of basis:

Let V be a finite dimensional with basis $B = \{e_1, \dots, e_n\}$. Every $v \in V$ is written uniquely as:

$$v = \sum_{i=1}^{n} a_i e_i = a_1 e_1 + \dots + a_n e_n$$

and $(a_1, \dots, a_n)$ are the coordinates of v in B.

Consider $S = \{e_1', \dots, e_m'\} \subset V$ expressed in B: $e_i' = a_{i1} e_1 + \dots + a_{in} e_i$. Let $A = (a_{ij})$ be the matrix of coefficients

Th. The dimension of the subspace $\langle S \rangle$ equals the rank of A:

$$\dim \langle S \rangle = \operatorname{rank} A$$

Obs. For $m = n$, $S = \{e_1', \dots, e_n'\}$ is a basis of $V \iff \operatorname{rank} A = n$

Def. The matrix A^T, denoted $P^{(e, e')}$, is the change of basis matrix from B to B':

$$P^{(e,e')} = \begin{pmatrix} a_{11} & a_{21} & \dots & a_{n1} \\ a_{21} & a_{22} & \dots & a_{n2} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

Its colluems are the coordinates of the new basis vectors e_j' in the old base e.

Properties:

1) Composition of basis changes: $P^{(e,e')} P^{(e',e'')} = P^{(e,e'')}$

2) $P^{(e,e')} P^{(e',e)} = I_n \iff (P^{(e,e')})^{-1} = P^{(e',e)}$

Th. Let $B = \{e_1, \dots, e_n\}$, $B' = \{e_1', \dots, e_n'\}$ be bases of V with change of basis matrix $P^{(e,e')}$. If $v \in V$ has coordinates $(v)_e$ in B and $(v)_{e'}$ in B', then:

$$(v)_{e'} = P^{(e',e)} \cdot (v)_e \iff (v)_e = P^{(e,e')} \cdot (v)_{e'}$$